

LeaveBalance Duplicate Key Error – Technical Documentation

Issue: MongoDB E11000 duplicate key error during staff leave allocation

1. Problem

While assigning leave balances to staff, MongoDB returned an E11000 duplicate key error similar to: **Plan executor error during findAndModify :: caused by :: E11000 duplicate key error**, with the conflicting key containing teacher: null, semester: ODD-SEMESTER, and academicYear: 2026-2027.

2. Root Cause

The LeaveBalance collection supports two ownership types: Teacher and Staff. A teacher balance uses the teacher field, while a staff balance uses the staff field. Staff documents can therefore legitimately have teacher set to null. The application schema had a partial unique index for teacher, but the actual MongoDB database contained an older teacher unique index without the partialFilterExpression.

Because the old teacher index was simply unique on teacher + semester + academicYear, multiple staff documents with teacher: null and the same semester/year were treated as duplicates.

3. Schema Design

The intended design is to maintain independent uniqueness rules for teachers and staff:

Owner	Unique combination	Condition
Teacher	teacher + semester + academicYear	teacher must be an ObjectId
Staff	staff + semester + academicYear	staff must be an ObjectId

4. Controller Behavior

The assignLeaves controller correctly distinguishes between teacher and staff targets. When target is staff, the query uses staff: user._id and \$setOnInsert also stores staff: user._id. When target is teacher, it uses teacher: user._id. Therefore, the controller logic was compatible with the intended index design.

```
const buildQuery = (userId) =>
  target === "teacher"
    ? { teacher: userId }
    : { staff: userId };

const buildOwner = (userId) =>
  target === "teacher"
    ? { teacher: userId }
    : { staff: userId };
```

5. Database Investigation

The original database indexes were inspected with:

```
db.leavebalances.getIndexes()
```

The teacher index was found to be unique but did not contain the required `partialFilterExpression`. The staff index already contained the correct partial filter.

6. Fix Applied

The outdated teacher index was removed:

```
db.leavebalances.dropIndex("teacher_1_semester_1_academicYear_1")
```

The correct partial unique teacher index was then created:

```
db.leavebalances.createIndex(
  {
    teacher: 1,
    semester: 1,
    academicYear: 1
  },
  {
    unique: true,
    partialFilterExpression: {
      teacher: { $type: "objectId" }
    }
  }
)
```

7. Final Index Configuration

The final database configuration was verified using:

```
db.leavebalances.getIndexes()
```

Index	Unique	Partial filter
<code>_id_</code>	Yes	—
<code>staff_1_semester_1_academicYear_1</code>	Yes	staff is ObjectId
<code>teacher_1_semester_1_academicYear_1</code>	Yes	teacher is ObjectId

8. Expected Behavior After the Fix

A staff LeaveBalance document can have `teacher: null` without participating in the teacher unique index. Different staff members can therefore receive balances for the same semester and academic year.

```
Staff A + ODD-SEMESTER + 2026-2027 -> allowed
Staff B + ODD-SEMESTER + 2026-2027 -> allowed
Staff A + ODD-SEMESTER + 2026-2027 -> updates existing record / prevents duplicate
```

Similarly, the teacher index continues to prevent duplicate balances for the same teacher, semester, and academic year.

9. Future Safety / Maintenance

The Mongoose schema and MongoDB indexes should remain synchronized. If an index definition is changed in the future, verify the actual indexes in MongoDB rather than assuming the existing database index has automatically changed.

Recommended verification command:

```
db.leavebalances.getIndexes()
```

Avoid removing the unique constraints unless the business requirement changes. The current unique constraints correctly enforce one balance per owner, semester, and academic year.

10. Conclusion

The issue was caused by a stale MongoDB teacher index. The application schema already defined the correct partial unique index, but the database still had the older non-partial unique index. Recreating the teacher index with `partialFilterExpression` resolved the collision between staff records whose teacher field is null.

Status: Fixed and verified.